Sami Ennedoui

Seeking a Work-Study Program (2.5 years): AI, Data, MLOps Engineer starting September 2026

Toulouse | +33 7 44 78 55 43 | sami.ennedoui@etu.toulouse-inp.fr

Useful Links (Portfolio, Work-Study Rhythm, GitHub...) *You can also scan the QR code→*

Profile

Engineering student in the ModIA double degree program (INSA/ENSEEIHT). I enjoy working on diverse topics spanning artificial intelligence, data processing, and embedded systems. My education has provided a strong foundation in applied mathematics and programming, and my projects have taught me to be rigorous, autonomous, and comfortable in team settings. I am looking for a work-study program where I can continue to learn, make tangible contributions to technical projects, and develop a broad vision of engineering.

Education

Toulouse INP–ENSEEIHT & INSA Toulouse 2025–2028
ModIA Double Degree (Modeling, Data, Artificial Intelligence) in Work-Study
Currently in 1st year of Electrical Engineering (3EA)
Key Courses: Probability & statistics, Linear algebra, Differential calculus and Optimization, Signal processing, C programming, MATLAB, Embedded Systems.

CPGE MP – Centre Tétouan 09/2023–06/2025
Preparatory Classes for Grandes Écoles. Solid foundations in analysis, linear algebra, probability, computational physics, and programming (Python, SQL). Intensive mathematical rigor in preparation for national competitive exams.

High School Diploma in Mathematical Sciences B (Engineering Sciences option) 09/2022–06/2023

Engineering & Modeling Projects

CHIP-8 Processor Design and Emulator 2026
— Implementation of an 8-bit retro interpreter coded in C (memory management, CPU cycles, graphics rendering).
— Learning and designing a RISC-V processor architecture on an FPGA board. *(personal projects)*

TIPE: Space-filling curves and image processing Presented 07/2025
— Mathematical proofs for spatial dimension reduction (2D → 1D).
— Python development of error quantification algorithms (Floyd-Steinberg, Atkinson).
— Lossless compression benchmarking: 89% size reduction with the Hilbert curve.

Professional Experience & Commitments

N7 Racing Team – BMS Pole Contributor (Embedded Systems) since 2025
— Sensor data acquisition and real-time safety monitoring of the battery pack (microcontroller boards).
— System architecture, coordination between powertrain and Data poles.

N7 Consulting (Junior-Enterprise) – Consultant Pole since 2025
— Technical consulting, client project management, and drafting deliverables.
— Translating complex technical requirements for non-technical stakeholders.

Skills

- **AI & Data Science** : Python (NumPy, SciPy, Scikit-Learn), MATLAB, Deep Learning, SQL.
- **Systems Programming** : C (low-level), C++, Bash, JavaScript (React).
- **Digital Electronics** : RISC-V, FPGA (Quartus, Vivado), SystemVerilog, CAN Bus, I2C.
- **Engineering & Tools** : Eclipse Capella (MBSE/Arcadia), Linux (Fedora), Git, System modeling.

Additional Training & Tech Watch

— **Deep Learning with Python Specialization (Coursera)** : Neural networks (ANN, CNN, RNN) *(in progress)*.
— **Project Management MOOC (Centrale Lille)** : Project management fundamentals *(in progress)*.
— Meetups & Events : AI-ficionados 3, Embedded Linux Meetup 9.

Languages

Arabic (**native**) | French (**fluent**) | English (**advanced**) | German (**beginner**)